

Quick Check

Lesson 6: Budding and Binary Fission: Making New Cells and Organisms from One Parent

7 multiple-choice questions • choose A, B, C or D

Sub-Strand 1.4: Cell Division | Lesson 6

A single yeast cell in dough can become millions of cells in a few hours. Which statement best explains how one yeast cell can produce a new organism?

- A** A male and a female yeast cell must be present
- B** The new organism arises from one parent cell without fusion of sex cells
- C** Two yeast cells must first fuse their sex cells
- D** The yeast cell must be fertilised by food such as sugar

A rod-shaped bacterium reproduces by binary fission. Which sequence correctly shows what happens?

- A** The cell splits in two, and only one daughter cell receives the chromosome
- B** The chromosome is copied, the cell lengthens, a dividing wall forms, and two equal cells result
- C** The bacterium fuses with another bacterium to form a new cell
- D** A small bud grows on the cell, receives a copy of the nucleus and then breaks off

One bacterium in contaminated water divides into two every 20 minutes. Assuming every cell keeps dividing, how many bacteria will there be after 2 hours?

A 64

B 32

C 6

D 12

Which statement is true for BOTH budding in yeast and binary fission in bacteria?

- A** Both copy the genetic material before the cell divides, and the new organism comes from one parent
- B** Both produce offspring that are genetically different from the parent
- C** Both always produce two daughter cells of equal size
- D** Both need the fusion of sex cells from two parents

A child grows from 3 kg to 50 kg by cell division. How is this similar to, but different from, a bacterium dividing by binary fission?

A

In both cases the cells divide, but in the child the new cells add to one body, while in a bacterium each new cell becomes a new organism

B

Only the child's cells copy their genetic material before dividing

C

Only the bacterium's division needs two parents

D

Both add cells to one body, so the cells stay joined together

Which is a correct difference between budding in yeast and binary fission in bacteria?

- A** Budding involves sex cells, while binary fission does not
- B** Budding happens in bacteria, while binary fission happens in yeast
- C** Budding needs two parents, while binary fission needs one parent
- D** Budding gives a smaller bud and a larger parent, while binary fission gives two equal cells

Bacteria that cause cholera and typhoid can multiply very quickly in unclean water. Why does boiling drinking water help prevent these diseases?

- A** Boiling turns the bacteria into yeast cells that are harmless
- B** Boiling allows the bacteria to fuse and become weaker
- C** Boiling kills the bacteria, so they cannot divide and increase in number
- D** Boiling makes the bacteria stop copying their DNA but they stay alive

All done

Ask your teacher how to submit your answers today (for example, on a paper answer slip).